FINITE-SAMPLE SAFE ADAPTIVE EARLY STOPPING FOR CORRELATED MAJORITY-VOTE ROLLOUTS

Anonymous authors
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ABSTRACT

Repeatedly sampling a model and taking a majority vote (self-consistency) is a standard way to spend inference compute, but the vote rarely needs all N samples. We give a *mixture-posterior stopper* (BAYES-H) whose per-problem conditional flip certificate is exact: given a calibration prior H over per-task pass counts, stop as soon as the posterior probability that the full-N majority will disagree with the current prefix majority drops below α. By the tower property the population flip rate of any such rule is bounded by α *without* a union bound over stopping times (adaptive is free), and the unknown-mixture layer adds only an empirical-Bernstein calibration step over problems, with Bonferroni control of the whole searched rule family. On 11,607 math problems with $N = 32$ rollouts each (OpenMathReasoning), selected and evaluated under a strict FIT/CAL/TEST problem split with a single held-out readout, BAYES-H has an 81% replay rollout-count reduction at realized flip rate $0.025 \leq \alpha = 0.05$ and 73% at $\alpha = 0.02$, dominating the strongest distribution-free baseline (Hoeffding-fixed-k: 16% at $\alpha = 0.05$) with a statistically significant paired gap, and the strongest mixture-aware fixed budget (empirical-Bernstein fixed-k: 47%). An ESC-style window heuristic with an 86% replay count reduction exceeds 0.05 ($0.058 > 0.05$). Under pre-fixed synthetic ordered-drift mechanisms (front-loading, linear trend, adversarial block swap), BAYES-H is the most robust stopper in the reported harness. The displayed certificate-margin thresholds are exploratory descriptive sensitivity results, not frozen-selection or confirmatory claims. The supplied replay readout replicates on a disjoint second shard; transferring one fitted calibration mixture across shards (TV 0.042) moves its reported quantities by ≤ 0.4 points. This is deterministic fitted-mixture sensitivity, not a population-transfer certificate. Two further model–carrier pairs bracket the mechanism: on OpenR1-Math-220k (DeepSeek-R1, $M = 2$ per problem) the certificate attains the replay risk–count frontier exactly when the mixture structure allows, and never stops under the stated replay certificate when it cannot; on RLVE (Qwen3-4B, $N = 8$ per problem, verifiable environments) it lands at the intermediate point the mechanism predicts — 51–58% replay count reduction at $\alpha \in \{0.10, 0.05\}$, 45–48% at $\alpha \in \{0.02, 0.01\}$, certified at *every* level down to $\alpha = 0.01$ where *no* distribution-free replay budget is certifiable. Every reported flip is disagreement between corresponding binary pass-count replay decisions, and every saving is $1 - \bar{k}/N$; these artifacts do not measure a delivered-answer majority, gold correctness, generated tokens, latency, cancellation, or ordered online deployment.

1 INTRODUCTION

Self-consistency (Wang et al., 2023) replaces one sampled answer by a majority vote over N samples. Adaptive variants reduce or reallocate sampling: ASC (Aggarwal et al., 2023) adapts a per-problem count with a lightweight stopping criterion, ESC (Li et al., 2024) proposes early stopping for self-consistency, and MARS (Chen et al., 2026) stops parallel traces using a conservative bound on future vote movement. For a count-only replay analysis, two deployment factors remain outside

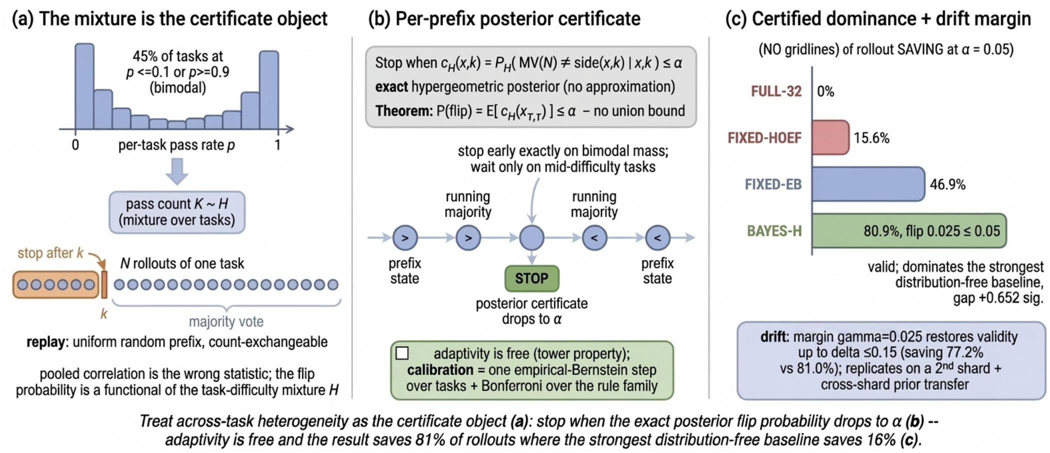

Figure 1: **Conceptual overview with artifact-checked quantitative labels.** (a) Treat across-task heterogeneity as the certificate object: per-task pass rates are strongly bimodal, so the flip probability is a functional of the difficulty mixture H, not of a pooled correlation. (b) BAYES-H stops when the exact posterior flip probability $c_H(x, k)$ drops to α; by the tower property adaptivity is free (no union bound), and calibration is one empirical-Bernstein step over tasks. (c) At $\alpha = 0.05$ BAYES-H has an 80.9% replay rollout-count reduction (realized flip 0.025), dominating the strongest distribution-free baseline (FIXED-HOEF, 15.6%) and the strongest mixture-aware fixed budget (FIXED-EB, 46.9%); a certificate margin controls flip only in stated synthetic drift mechanisms, and all results replicate on a disjoint second shard. Panel (c)'s displayed numbers are rounded reproductions of the named replay artifacts, not measurements generated by the figure. The flip and count quantities are replay artifacts, not online correctness or operational-cost measurements (REPRO_README.md).

its evidence: (i) within-task rollouts may be correlated; and (ii) chronological generation order may drift, breaking an exchangeability-based certificate.

We make two structural observations that reframe the problem. First, on real carriers the dominant deviation from i.i.d. is not latent within-task correlation but *across-task heterogeneity*: per-task pass rates are strongly bimodal (Section 3), and the flip probability a stopper must control is a functional of the *mixture* of per-task difficulty, not of a pooled correlation coefficient. Pooled second-moment summaries can be simultaneously over-conservative and uncalibrated (Section 3). Second, once the target is the mixture, the correct object is a *per-prefix conditional certificate*: the posterior probability, given the observed prefix, that the full-N majority disagrees. This object is exactly computable (hypergeometric posterior), its fixed-budget expectation $\mathbb{E}_H[f_K(k)]$ is monotone in the prefix length (Lemma 1), and — because the tower property collapses the stopping-time dependence — it certifies *any* adaptive rule at level α with no union bound (Theorem 1).

Contributions.

1. **Framework.** A count-exchangeable replay model in which both within-task correlation and across-task heterogeneity are folded into a mixture H over per-task pass counts; the certificate target is the flip probability under H, unifying the two effects (Section 3).
2. **Method.** BAYES-H: stop when the posterior conditional flip probability $c_H(x, k) \le \alpha$. Exact certificates (Lemma 1, Theorem 1); adaptivity is free — no union bound, no e-process tuning (Section 4).
3. **Legitimate selection.** A FIT/CAL/TEST problem-level protocol in which the rule family (budget $\times$ level for fixed-k; level for BAYES-H) is selected on CAL with simultaneous empirical-Bernstein bounds (Maurer & Pontil, 2009) plus Bonferroni, and TEST is read once (Section 5).
4. **Empirical replay result.** On 11,607 OpenMathReasoning problems, 80.9% replay rollout-count reduction at realized flip 0.025 ($\alpha = 0.05$), vs 15.6% for the strongest distribution-free baseline at the same level; ESC-style window stopping is invalid ($0.058 > 0.05$). Gains are paired-CI significant (Section 6). Two further carriers with different models (DeepSeek-R1 at

$M = 2$; Qwen3-4B at $N = 8$ on verifiable environments) show the certificate attains the replay risk–count frontier (stop when the replay certificate permits, otherwise not) and interpolates with the resolvable mixture structure, i.e. the method is carrier-aware rather than a generic stop-early trick (Section 6).

5. **Honest boundaries.** The public carrier gives per-task pass *counts*, not ordered rollouts; it supports a random-prefix (count-exchangeable) analysis only as a model assumption and cannot validate ordered online deployment. We therefore bound the failure domain under three pre-fixed synthetic ordered-drift mechanisms; displayed margin thresholds are exploratory/descriptive (Section 7).

2 Related work

Adaptive sampling for voting. ASC (Aggarwal et al., 2023) adapts a per-problem sample count using a lightweight stopping criterion, and ESC (Li et al., 2024) proposes early stopping for self-consistency. We do not establish a finite-sample disagreement-with-full-vote guarantee for either; on our supplied heterogeneous replay carrier, the instantiated ESC-style rule exceeds 0.05 (Table 1). MARS (Chen et al., 2026) is a parallel trace-stopping method that bounds future vote movement; our work is instead a sequential count-exchangeable replay certificate. Best-of-n and budget-forcing methods (Snell et al., 2025) tune compute without stopping certificates.

Conformal and anytime inference. Conformal prediction (Vovk et al., 2005) gives marginal coverage under exchangeability; confidence sequences (Howard et al., 2021) give anytime-valid bounds for i.i.d. means. Our certificate is neither: it is an *exact posterior flip probability* under a calibrated discrete prior, valid conditionally per prefix (hence adaptivity is free), with the mixture estimated from calibration problems under an empirical-Bernstein bound (Maurer & Pontil, 2009). The Bonferroni control of a small pre-declared rule family is deliberately conservative and transparent.

Dependent sampling. Design-effect corrections (Kish, 1965) shrink effective sample size by an intraclass correlation; we show pooled correlation is the wrong functional here (Section 3). Beta-binomial and correlated-Bernoulli models (Skellam, 1948) capture overdispersion but target estimation, not stopping certificates for the flip event.

Feature boundary. Table 2 makes the source-verified distinction compact: the proposed object is a specified count-exchangeable full-vote *replay* endpoint with a calibrated-mixture posterior flip certificate. It is not a general ordered-online early-stopping, conformal, or anytime-inference guarantee.

3 Model: count-exchangeable replay

A task's N rollout outcomes are summarized by the pass count $K \in \{0, \dots, N\}$. Across tasks, $K \sim H$, a mixture capturing both task heterogeneity and any within-task correlation: the probability a fresh rollout passes, marginally, is $\mathbb{E}_H[K]/N$, and the joint law of the N outcomes is exchangeable *within* a task given its realized count. Analysis model (*replay*): the stopper observes a uniform random prefix drawn without replacement from the task's N outcomes — i.e., the first k draws are exchangeable with the remaining $N - k$. Section 7 stress-tests departures from this model (ordered drift); our validity claims are stated relative to it, and the carrier limitation (counts only, no order) is made explicit in Section 8.

Let $\mathrm{MV}(k) = \mathbf{1}\{x_k > k/2\}$ with ties assigned to the negative side, and let $\mathrm{MV}(N) = \mathbf{1}\{K > N/2\}$ be the full-N *binary pass-count replay decision*. It is not an aggregation of delivered textual answers or an actual delivered answer-majority: the replay model contains only binary pass/fail outcomes. The *flip event* of a stopping rule τ is $\{\mathrm{MV}(\tau) \neq \mathrm{MV}(N)\}$.

Remark 1 (Why the mixture, not the correlation). On τ^2-bench carriers (6 public task suites, 4 rollouts/task) we measured pooled within-task reward correlation $\hat{\rho} = 0.47$–0.59, but a task-demeaned (fixed-effect) estimate gives $\hat{\rho}_{\mathrm{FE}} \approx -1/3 \approx -1/(N-1)$, the demeaning artifact floor: essentially *all* observed correlation is across-task heterogeneity, not latent within-task dependence. On OpenMathReasoning the per-task pass-rate variance exceeds the binomial baseline by $34\times$. The flip probability is driven by the mass of tasks with $K \approx N/2$ — a mixture functional — so pooled second-moment discounts are the wrong certificate statistic.

4 METHOD: THE MIXTURE-POSTERIOR STOPPER

Exact per-task flip probability. Given count K, the prefix count x_k is hypergeometric $\mathrm{HG}(N, K, k)$, hence

$$f_K(k) \ := \ \mathbb{P}\big(\mathrm{MV}(k) \neq \mathrm{MV}(N) \,\big|\, K\big) = \sum_x \mathrm{HG}(N, K, k, x)\,\big|\mathrm{side}(x, k) - \mathrm{side}(K, N)\big|. \quad (1)$$

Lemma 1 (Exactness and monotonicity)**.** (1) *is an identity (no approximation). For every fixed K, $f_K(k)$ is nonincreasing in odd k; hence for any mixture H, $\mathbb{E}_H[f_K(k)]$ is nonincreasing and $k^*(\alpha) = \min\{k : \mathbb{E}_H[f_K(k)] \le \alpha\}$ is well defined and binary-searchable.*

Proof. The identity is a definition expansion. Monotonicity follows from a coupling of the prefixes at k and $k{+}2$ on a common ordered draw. The proof first handles the support-boundary case where the downward boundary atom has probability zero directly; only otherwise does it divide to obtain the ratio $(K-t)/(N-K-t) \ge 0$ for $K > N/2$. The complementary case follows by pass/fail reflection (Appendix A). An exact-rational CPU audit checks every admissible (N, K, k) for $2 \le N \le 64$, including $(32, 17, 29)$. □

Conditional certificate. Given prior H and an observed prefix (x, k), define the posterior flip probability

$$c_H(x, k) \ := \ \mathbb{P}_H\big(\mathrm{side}(K, N) \neq \mathrm{side}(x, k) \,\big|\, x, k\big), \qquad \text{posterior} \propto H[K] \cdot \mathrm{HG}(N, K, k, x). \ (2)$$

We define the terminal certificate by $c_H(x, N) := 0$. At a posterior-supported terminal state, the hypergeometric likelihood has support only on $K = x_N$, so this convention equals the conditional flip probability: $c_H(x_N, N) = 0$. It also gives the correct endpoint value for a zero-prior-mass row of a frozen certificate table, because the observed total count already fixes the binary full-N replay decision.

Theorem 1 (Adaptive is free)**.** *Let τ be any stopping time measurable w.r.t. the prefix filtration that stops only at states with $c_H(x_\tau, \tau) \le \alpha$. The terminal state $k = N$ satisfies this same rule because $c_H(x_N, N) = 0$. Then $\mathbb{P}(\mathrm{MV}(\tau) \neq \mathrm{MV}(N)) = \mathbb{E}[c_H(x_\tau, \tau)] \le \alpha$, with equality in the identity and no union bound over stopping times.*

Proof. $\mathbb{P}(\text{flip}) = \mathbb{E}[\mathbb{P}(\text{flip} \mid \text{prefix path})]$. For any path stopped at (x, k), $\mathbb{P}(\mathrm{MV}(N) \neq \mathrm{side}(x, k) \mid \text{path}) = c_H(x, k)$ since the posterior is exactly the conditional law of K and $\mathrm{MV}(N)$ is a function of K. At $\tau = N$, $x_N = K$ and hence this conditional probability is $c_H(x_N, N) = 0$; at every earlier stopped state it is at most α by the rule. Thus the integrand is bounded pointwise by α, and taking expectations gives the claim. □

BAYES-H is the rule $\tau = \min\{k \ge 3 : c_H(x_k, k) \le \alpha\}$ with $H = \widehat{H}_m$, the calibration empirical mixture. Its per-task exact flip probability and expected stopping time under any true count K are computable by a linear-time DP over prefix states (k, x) (Appendix B), enabling exact calibration and evaluation without Monte Carlo.

Theorem 2 (Population layer)**.** *Let $g(K; \alpha, \widehat{H}_m)$ be the exact per-task flip probability of BAYES-H under count K, and let $\bar{g}_{\mathrm{CAL}}$, s^2_{CAL} be its mean and variance over m calibration problems. With probability $1-\delta$ over the calibration sample,*

$$\mathbb{P}_{\mathit{replay},H}(\text{flip}) \ \le \ \bar{g}_{\mathrm{CAL}} + \sqrt{\tfrac{2 s^2_{\mathrm{CAL}} \log(4/\delta)}{m}} + \tfrac{7 \log(4/\delta)}{3(m-1)},$$

by the empirical-Bernstein inequality (Maurer & Pontil, 2009) applied to the bounded per-task flips $g \in [0, 1]$, with δ split by Bonferroni over the pre-declared selection family ($J = 64$ candidates; Section 5).

Remark 2 (No Jensen gap). The population flip rate under replay is a *linear* functional of H: $\mathbb{P}(\text{flip}) = \sum_K H[K]\, g(K)$. Hence estimation error in $\widehat{H}_m$ enters only through the empirical mean of a bounded per-task quantity — there is no Jensen correction, and empirical Bernstein is essentially tight at $m = 4000$.

Proposition 1 (Exact deterministic TV sensitivity of a frozen rule)**.** *Fix the rule (certificate tables frozen from $\widehat{H}$) and let $g(K)$ be its exact per-task flip probability. Over the total-variation ball $B_R = \{q : q \geq 0, \sum_K q_K = 1, \|q - \widehat{H}\|_1 \leq 2R\}$,*

$$V(R) \;:=\; \max_{q \in B_R} \; \sum_K q_K\, g(K)$$

is a linear program whose optimum is a two-sided greedy: add mass to atoms in descending g (capped by $q_K \leq 1$) and drain mass from atoms in ascending g (capped by $\widehat{H}_K$); $V(R)$ is hence computable in closed form in $O(N \log N)$. For the positive radii and realized capacities checked in the supplied LP grids, it is strictly below the 1-Lipschitz envelope $\text{flip}(\widehat{H}) + R$*; no universal strict-refinement claim is made. The* critical radius $\tau^*(\alpha) = \max\{R : V(R) \leq \alpha\}$ *is a deterministic property of the frozen rule and fitted reference mixture. It becomes a population certificate only after separately proving that an unknown target mixture lies in B_R, which current artifacts do not do.*

On the OMR carrier the critical radii are $\tau^*(0.10/0.05/0.02/0.01) = 0.144/0.078/0.074/0.052$ (shard 0) and $0.157/0.082/0.076/0.024$ (shard 1) (`tv_robustness_r481_result.json`; the closed form matches a direct simplex LP to 10^{-6} on a $6 \times 4 \times 2$ grid of radii, levels and shards). Three consequences: *(i)* at the observed cross-shard mismatch $R = 0.042$, the fixed-rule sensitivity calculation returns 0.039 at the 0.05 table and 0.016 at the 0.02 table. It describes an assumed ball around a fitted mixture, not containment of either shard's unknown population mixture; at 0.01, $\tau^*(0.01) = 0.024 < 0.042$. *(ii)* Tightening a frozen table gives a deterministic B_R sensitivity calculation (at $R = 0.05$, the 0.10 table re-evaluated at 0.05 has $V(R) = 0.042$ and 80.3% replay count reduction), not a "validity band" without a target-mixture confidence set; *(iii)* at every level a 36–47% mass of atoms has $g(K) = 0$ exactly (the certificate stops only on states the prefix already decides), which is why $V(R)$ grows far slower than the Lipschitz rate $R \cdot \max_K g(K)$. *Cross-carrier.* On the RLVE carrier (N=8; same frozen FIT prior as Section 6, no TEST re-read) the same closed form gives $\tau^*(0.10/0.05/0.02/0.01) = 0.199/0.551/0.213/0.449$ (`tv_robustness_rlve_r482_result.json`; LP cross-check matches to 10^{-6}). The radii are *larger* than on OMR at every level even though the RLVE prior is no more heterogeneous (het-excess variance 0.0999 vs 0.129): the RLVE prior concentrates 74–87% of its mass on *zero-g* atoms (vs 36–47% on OMR), and with only N=8 atoms the worst-case adversary has few high-g directions in which to move mass. Across carriers τ^* co-varies with the *alignment* of the prior with the zero-g set and is uncorrelated with heterogeneity (het-excess vs τ^*: Spearman 0.01; het-excess vs alignment: 0.05 — the two candidate drivers are not confounded). The alignment–τ^* co-variation is a *between-carrier* law: pooled Spearman 0.75 over the 16 carrier$\times$level cells, 1.0 over the four carrier means, 0.63 with the two whole-simplex-plateau cells removed; *within* OMR it is negative (-0.95/-0.89 per shard — tightening the level moves the zero-g set only at the most-aggressive end, where mass is added on the lowest-g atoms), positive within RLVE (0.32) and OpenR1 (1.0) (decomposition audit: `spearman_decomposition_r486_result.json`; Appendix E). (Two disclosures: τ^* bisection uses $R \in [0, 1]$ here ($0.551 > 0.5$ exceeds the $[0, 0.5]$ range sufficient for OMR); τ^* is non-monotone in α — $\tau^*(0.05) > \tau^*(0.10)$ — because tightening α zeroes K=6 ($g : 0.107 \to 0$) and shrinks K=5 ($0.339 \to 0.071$), flattening the g-profile's top faster than the budget tightens; internal drops exist as well (the curve has 7 rule-change breakpoints on the plotted range, Figure 2), so τ^* should not be read as globally monotone in α.) *Three-atom edge case.* On OpenR1 (M=2; same frozen FIT prior, no TEST re-read) the support is three atoms $p \in \{0, \frac{1}{2}, 1\}$ and $g(0) = g(1) = 0$ at every level (a decided prefix cannot be overturned by one more rollout), so the worst case is a pure one-sided pump into the unique positive-g atom: $V(R) = \text{flip}(\widehat{H}) + g_{\max} R$ exactly and $\tau^*(\alpha) = \alpha/g_{\max} - \widehat{H}(\frac{1}{2})$ in closed form (`tv_robustness_openr1_r483_result.json`; scan match 10^{-9}, LP 10^{-6}). Since $g_{\max}$ tracks the certificates ($0.0785/0.0460$): $\tau^*(0.10) = \tau^*(0.05) = 0.168$ (halving α past 0.0785 halves $g_{\max}$, $0.25 \to 0.125$), while $\tau^*(0.02) = \tau^*(0.01) = 1$ — below the smallest certificate the frozen rule never stops, $V(R) \equiv 0$, no prior shift can break it. Non-monotonicity is thus maximal here ($0.168 \to 1$); zero-g mass 76.8% sits between OMR's 36–47% and RLVE's 74–87%, completing the pattern from the low-support side. *Conservation curve.* Appendix E maps $\tau^*(\alpha)$ on the dense continuum $\alpha \in [10^{-3}, 0.2]$ for all three carriers: the curve is analytically staircase-shaped — non-decreasing inside each constant-rule interval, so that every jump of τ^* is a strict *drop* exactly at a certificate threshold (analytic on OpenR1: τ^* drops by 0.314 at $\alpha = 0.0785$ and by 0.864 at 0.0460). The practical reading is a staircase interpretation of the certificate level: at a rule-change breakpoint, a small budget relaxation buys a discrete jump in certified drift tolerance, so α should be set on a step, not inside one.

Remark 3 (Continuous-α selection is free for BAYES-H). The stopping condition $c_{\widehat{H}}(x,k) \le \alpha$ does not involve α on the left-hand side, so the BAYES-H rule family is *nested* in α: as α varies continuously, the rule changes only at the finitely many breakpoints given by the distinct values of $c_{\widehat{H}}(x,k)$ — at most $\sum_{k=3}^{N}(k+1)$ stoppable states (555 at N=32; 267 distinct values observed, plus the never-stop rule). These are three distinct counts: 555 stoppable states bound the possible rule changes; only 267 *distinct* certificate values are observed on the support, and the rule family has one rule per value *plus* the never-stop rule (268 in total; shrunk to 182 below for the tested range). A union bound over this breakpoint family therefore certifies the *entire continuum* $\alpha \in (0, 0.10]$ simultaneously — no pre-declared α grid is needed. The family can be shrunk further with no loss: for the range $\alpha \in (0, 0.10]$ only the 181 distinct positive certificate values ≤ 0.10 are rule-distinguishing (values > 0.10 never trigger any rule in the range; zero-certificate states stop under every $\alpha > 0$), so the exact effective family has 182 rules ($J = 6{,}501$ including the FIXED-k part) instead of the 9,339 obtained by counting all observed breakpoints; the four reference selections below are unchanged and their UCBs tighten by $\le 6 \times 10^{-4}$ (`prop_562bound_r479_result.json`). On our calibration pool the enlarged family ($J_{\text{cont}} = 9{,}339$ vs. $J = 64$) inflates UCBs only through the logarithmic factor; the selected replay-count-optimal BAYES-H rule is bit-for-bit identical at $\alpha \in \{0.10, 0.05, 0.02\}$ and tightens only at $\alpha = 0.01$ (mean k $10.3 \to 11.9$), while the weaker FIXED-EB baseline loses its $\alpha = 0.02$ budget under *either* family size (`alpha_continuous_r478_result.json`).

5 PROTOCOL: FIT/CAL/TEST WITH A LEGITIMATE SELECTION FAMILY

Split. 11,607 problems are partitioned once (seeded) into FIT (4000), CAL (4000), TEST (3607). FIT estimates $\widehat{H}$ and freezes the BAYES-H certificate tables. CAL selects rules from the pre-declared family: fixed-k candidates $k \in \{3, 5, \ldots, 31\} \times \alpha \in \{0.10, 0.05, 0.02, 0.01\}$ (60 certificates) and BAYES-H at the same 4 levels, with simultaneous empirical-Bernstein UCBs at per-test $\delta = 0.05/64$. $k^*(\alpha)$ is the smallest budget whose CAL UCB is $\le \alpha$; BAYES-H is accepted at level α iff its CAL UCB is $\le \alpha$. TEST is read exactly once. All evaluation uses the exact per-task flip functions ((1) for fixed-k; the DP of Appendix B for BAYES-H), so TEST numbers are exact replay quantities, not MC estimates; the DP itself is cross-checked against replay MC (max deviation 0.043 within $3\sigma_{\text{MC}} = 0.075$).

Baselines. FULL-N; FIXED-HOEF (fixed budget selected with distribution-free Hoeffding UCBs, the strongest model-free competitor); FIXED-EB (fixed budget with the same mixture-aware empirical-Bernstein machinery as BAYES-H — isolates the value of adaptivity); BAYES-UNIF (uniform prior; not a certificate when the prior is wrong); WINDOW3 (ESC-style: stop when 3 consecutive samples agree with the current majority — no certificate knob).

6 EXPERIMENTS

Carrier. OpenMathReasoning CoT shard 0 (Moshkov et al., 2025) (CC-BY-4.0): the `pass_rate_72b_tir` column gives, for 11,607 unique problems, the exact pass count K of Qwen2.5-Math-72B-TIR over $N = 32$ rollouts — no answer-extraction noise. Pass rates are strongly bimodal: 45% of problems have $p \le 0.1$ or $p \ge 0.9$; 11.6% sit in $[0.4, 0.6]$. This carrier is a count-only carrier used for the *count-exchangeable replay* analysis: it supplies per-task counts but no rollout order, so the random-prefix assumption is a modeling condition rather than an observed chronology (Section 7).

Main table. Table 1 reports the single TEST readout (Bonferroni CI radius 0.0286 across the 9 reported rules). In this section, "valid" means replay full-vote flip $\le \alpha$; TEST is descriptive (selection was certified on CAL). Neither flip nor count reduction is gold correctness or an operational token, latency, or cancellation measurement.

Table 1: TEST readout ($n = 3607$ problems, $N = 32$; 8 rule–level rows under the Bonferroni CI of Section 5 plus FULL-32 as the zero-count-reduction reference). Replay count reduction $= 1-\bar{k}/32$. BAYES-H attains the largest valid replay count reduction at $\alpha = 0.05$ and 0.02; at $\alpha = 0.10$ FIXED-EB has 84.4% replay count reduction versus BAYES-H 84.0% (a 0.4-point gap in mean replay count on this test pool, descriptive, no paired CI computed for this pair), while BAYES-H keeps the lowest realized flip at every level. WINDOW3 has the largest replay count reduction but is invalid at $\alpha \leq 0.05$.

Rule	α	replay full-vote flip	mean k	count reduction
FULL-32	—	0	32.0	0
WINDOW3 (ESC-style)	—	**0.0577**	4.4	86.2%
FIXED-HOEF k^*=7	0.10	0.0645	7.0	78.1%
FIXED-EB k^*=5	0.10	0.0794	5.0	84.4%
BAYES-H	0.10	**0.0370**	5.1	**84.0%**
FIXED-HOEF k^*=27	0.05	0.0156	27.0	15.6%
FIXED-EB k^*=17	0.05	0.0324	17.0	46.9%
BAYES-H	0.05	**0.0249**	6.1	**80.9%**
FIXED-HOEF	0.02	*no certifiable budget*		
FIXED-EB k^*=31	0.02	0.0083	31.0	3.1%
BAYES-H	0.02	**0.0091**	8.5	**73.5%**

Fair comparison. Against the strongest distribution-free baseline (FIXED-HOEF) at the same α, BAYES-H's paired per-problem replay-count-reduction gap is $+0.058$ (CI radius 0.029) at $\alpha = 0.10$ and $+0.652 \pm 0.029$ at $\alpha = 0.05$ — a $5.2\times$ improvement over the baseline's 15.6% count reduction; at $\alpha = 0.02$ the baseline has *no* certifiable budget while BAYES-H still has a 73.5% reduction. Against the strongest mixture-aware fixed budget (FIXED-EB), adaptivity adds $+34$ points of replay count reduction at $\alpha = 0.05$ (80.9% vs 46.9%) and $+70$ points at $\alpha = 0.02$ — the adaptive advantage grows as α tightens, because the conditional certificate stops early exactly on the bimodal mass (extreme-K problems stop at $k \approx 5$) and waits only on the 11.6% of mid-difficulty problems.

Cross-source transfer. Splitting CAL/TEST across the four AoPS problem sources (train on three, test on the fourth), BAYES-H stays valid on all transfers (replay flip 0.021–$0.033 < \alpha = 0.05$; count reduction 79–83%). This supplied count-replay readout is descriptive of these source splits; it does not establish ordered online transfer.

Second-shard replication and prior-mismatch transfer. We replicate the entire chain on a disjoint carrier, OpenMathReasoning CoT shard 1 (11,454 problems, same N=32 pass counts; SHA-256 pinned), whose mixture structure is near-identical (heterogeneity excess 0.125 vs 0.129; 44.4% extreme-p mass). Two runs share the CAL/TEST problems of shard 1 and differ only in the prior's provenance. *(i) Within-shard*: the prior is fit on shard 1's own FIT split; every qualitative replay conclusion reproduces — BAYES-H has 80.6% replay count reduction at realized flip 0.025 ($\alpha = 0.05$) and 73.1% at 0.009 ($\alpha = 0.02$), vs 15.6% and *no certifiable budget* for FIXED-HOEF. *(ii) Cross-shard prior transfer*: the prior is fit on shard 0 FIT (fitted-mixture TV distance to the shard 1 fit $= 0.042$) and the certificate is selected and read on shard 1. All numbers move by ≤ 0.4 points (count reduction 80.6%/73.2%; flips 0.025/0.009). This is a small movement in one supplied count-replay transfer and a deterministic fitted-mixture sensitivity diagnostic, not a statistical population-transfer certificate or a demonstration of robustness to naturally ordered online rollouts (Section 3).

Second model–carrier pair: OpenR1-Math-220k (DeepSeek-R1, $M = 2$). To probe the mechanism under a different model and a structurally opposite carrier, we run the identical chain on OpenR1-Math-220k (MIT; 9,374 unique problems, two deduplicated DeepSeek-R1 rollouts each, HF snapshot `e4e141ec`). Under the replay model, the full vote is the 2-vote majority (ties broken by a pre-drawn fair coin), the only stoppable prefix is $k = 1$, and per-problem risk is analytic. The only identifiable prior at $M = 2$ is coarse (three atoms $p \in \{0, \frac{1}{2}, 1\}$, fit by exact pair-type moment matching): 25% always-fail, 23% coin-flip, 52% always-pass. Results (FIT/CAL/TEST 3000/3000/2853): *(i)* the conditional certificate attains the replay risk–count *frontier*: at $\alpha = 0.20$ it stops on every problem (flip $0.059 \le \alpha$, 50% replay count reduction); at $\alpha = 0.10$ full stopping has the tightest distribution-free UCB $0.080 \le \alpha$ and is certified with the same realized flip and count reduction; at $\alpha = 0.05$ it stops only on the sub-population whose observed prefix certifies (positive-side prefix), giving 32% replay count reduction at flip $0.029 \le \alpha$; at $\alpha = 0.02$ it *never* stops — and no legal certificate can, since stopping at $k{=}1$ on any problem carries posterior risk > 0.02; *(ii)* the realized flip of always stopping at $k{=}1$ (0.059) matches its analytic value $\mathbb{E}_H[p(1{-}p)] = 0.058$ to Monte-Carlo precision. Replay count reduction is bought by mixture *structure* that a prefix can resolve: with $M = 2$ there is at most one draw of prefix information, and the certificate extracts exactly the count reduction that structure allows — 50% at loose levels, partial at $\alpha = 0.05$, honestly nothing at $\alpha = 0.02$ — versus 81%/73% at $\alpha = 0.05/0.02$ on the $N{=}32$ OMR carrier. This is cross-model replay evidence for the stated mechanism; it does not test chronological online deployment, gold correctness, or operational cost.

Third model–carrier pair: RLVE (Qwen3-4B, $N = 8$). The two carriers above close the extremes ($N{=}32$ rich-prefix vs $M{=}2$ near-prefix-free). To probe the intermediate regime under a third model family and a different task family, we run the identical chain on RLVE teacher rollouts (Qwen3-4B-Thinking-2507, pass@8 on 9,000 verifiable counting/combinatorics environments; Apache-2.0; HF snapshot `eaeec946`). Success is the offline Gym-verifier reward being positive (a small mass of fractional partial credit is counted as failure). The mixture is again strongly U-shaped (42% extreme problems, heterogeneity excess 0.100 over the binomial baseline). Results (FIT/CAL/TEST 3000/3000/3000): *(i)* BAYES-H has replay count reduction 57.6%/50.9%/48.2%/45.4% at $\alpha = 0.10/0.05/0.02/0.01$ with realized flips 0.031/0.010/0.005/0.002 — conservative at every level; *(ii)* the certificate passes at *every* level down to $\alpha = 0.01$, while *no* distribution-free budget is certifiable below $\alpha = 0.10$ (FIXED-EB and FIXED-Hoeffding both null at $\alpha \le 0.05$) — at $\alpha = 0.10$ the paired replay-count-reduction gap over FIXED-Hoeffding is +0.20 (CI radius 0.03, significant); *(iii)* the exact DP prediction matches a 3,000-draw replay Monte-Carlo on held-out problems (flip within 0.004, mean k within 0.04). Two descriptive diagnostics: the recorded sample-id success slope is flat (pooled slope -7.8×10^{-5}/trial; per-position rates 0.316–0.326), consistent with near-exchangeable sampling in this carrier; and the replay count reduction interpolates with resolvable structure exactly as the mechanism predicts (81% at $N{=}32$ with 15 stoppable budgets, 45–58% at $N{=}8$ with 3, 32–50% at $M{=}2$ with 1; all in-distribution replay readouts collected in Appendix F). The sample-id diagnostic does not audit an online policy or make the count-replay certificate valid for deployment order.

Why the window heuristic fails. On mid-difficulty problems ($K \approx 16$), three consecutive agreements with the current majority is a *common* event under near-coin-flip draws, not evidence the vote has settled: WINDOW3 realizes flip 0.058 at mean $k = 4.4$. Its flip rate is nearly flat in α (it has no certificate knob): it can only be tuned by window length, and its certified population UCB is 0.061 (`adaptive_sweep_r468_result.json`), so it barely misses even the loosest tested level $\alpha = 0.10$ on the full OMR test pool.

7 SYNTHETIC ORDERED DRIFT: FAILURE DOMAIN AND MARGIN REPAIR

The replay model assumes prefixes are exchangeable with the held remainder. Because the carrier contains no observed chronology, we apply three *pre-fixed synthetic* drift mechanisms to its true counts ($\delta \in \{0, 0.05, 0.10, 0.15, 0.20\}$, 2000 TEST problems, 500 orders each): **E1** (front-load: the first k draws are enriched/depleted in passes by count $D = \lceil 2\delta \min(K, N{-}K) \rceil$), **E2** (linear per-draw pass-rate trend $p_j = K/N + \delta(2j/(N{-}1){-}1)$), **E3** (adversarial block: the last $\lceil \delta N \rceil$ draws drift toward the minority side).

Findings (full tables in Appendix D): (i) all methods break under E1 by $\delta = 0.10$–0.15 in this harness; the certificate is honest about its premise — E1 violates exchangeability by construction; (ii) under the gentler E2/E3, BAYES-H is the *most* robust stopper in this synthetic grid: at $\alpha = 0.10$ it remains valid to $\delta = 0.20$ under E2 (flip 0.083) while FIXED-k breaks at $\delta = 0.15$ and WINDOW3 breaks at $\delta = 0.20$ (flip 0.104; under E3 WINDOW3 stays valid throughout, its last-window rule never reaching the drifting tail); at $\alpha = 0.05$ under E3 it breaks only at $\delta = 0.15$ (flip 0.052), while FIXED-EB survives (it stops at $k = 17$, having seen half the block); (iii) an exploratory margin scan illustrates a synthetic flip/count tradeoff. No frozen margin-selection record is bundled, so its thresholds are not confirmatory. Under E3 at $\alpha = 0.05$, the explored margin $\gamma = 0.025$ covers $\delta \leq 0.15$ (flip 0.0495, replay count reduction 77.2% vs 81.0% unmargined in the stress harness; the unmargined exact TEST readout of Table 1 is 80.9%); covering the extreme $\delta = 0.20$ adversarial block requires $\gamma > 0.03$ (count-reduction cost > 7 points), and under the smoother E2 trend $\gamma = 0.03$ covers the full tested range $\delta \leq 0.20$ (flip 0.0491, count reduction 71.8%). The drift budget δ could be estimated on CAL only if ordered rollouts are retained; on this count-only carrier it is assumed. This synthetic analysis neither measures nor certifies natural online order.

8 LIMITATIONS

(i) The public carrier provides pass *counts*, not ordered rollouts: all main results are exact only under the count-exchangeable replay model; the synthetic stress mechanisms of Section 7 only characterize specified departures from that model. The certificate is a replay-model conditional certificate, not an e-process: validity under arbitrary online stopping times in ordered deployment is outside the stated claims. No supplied run compares the stopped answer with gold correctness, or records generated tokens, latency, cancellation acknowledgement, or post-cancellation work; full-vote agreement and $1-\bar{k}/N$ cannot substitute for those endpoints. (ii) The mixture prior is estimated from 4000 calibration problems of the same distribution; adversarial mixture shift beyond the tested transfer range can break the certificate; the exact critical radii of Proposition 1 give deterministic frozen-rule sensitivity radii over an assumed fitted-mixture TV ball ($\tau^* = 0.024$–0.157 across levels and shards); they do not establish target-population containment. Within that assumed ball the failure boundary is sharp — at $\alpha = 0.01$ on shard 1 the observed cross-shard mismatch 0.042 already exceeds $\tau^*(0.01) = 0.024$; and the radii are *rule-conditional*: $\tau^*(\alpha)$ is a staircase with jumps at the certificate thresholds (Appendix E), so the level α should be read as choosing a step of that staircase, not a point of a smooth trade-off. Along the estimation axis, Appendix E(e) shows τ^* is not monotone in the prior-fit size m; the decomposition of Appendix E(f) attributes this to the *rule* the fit induces (freezing the prior leaves the non-monotonicity intact; freezing the rule removes the largest dips), not the prior's sampling error. Appendix E(g) turns that attribution into a constructive repair: a per-atom flip budget implemented as a certificate-ordered subset of stopping states provably dominates the induced profile, restores the full simplex ($\tau^* = 1$, trivial frozen-rule regime) at 70 of 72 cells for $+1\%$–$+11\%$ extra votes, and repairs the coarse-fit breaks below — while two alternatives (prior support truncation, deadline truncation) provably do not. On OMR shard 1 the coarsest fit $m{=}125$ breaks the transferred certificate at $\alpha \leq 0.02$ while $m \geq 250$ — or the flip-budget repair at any m — restores it. Appendix F collects every in-distribution readout and its failure boundary in one matrix. (iii) Lemma 1's monotonicity is proved in closed form (Appendix A) and independently verified exhaustively at $N = 32$. (iv) Main-table counts come from one model–task pair (Qwen2.5-Math-72B-TIR, math problems); generality across problem pools is supported by the disjoint second-shard replication and cross-shard prior transfer, and generality across models and task families by two further model–carrier pairs — DeepSeek-R1 rollouts on OpenR1-Math-220k ($M = 2$), on which the certificate attains its replay risk–count frontier, and Qwen3-4B rollouts on RLVE verifiable environments ($N = 8$), on which it is certified at every tested level and interpolates with the resolvable mixture structure (Section 6). (v) At $k = N$, the observed pass count equals K, so the binary pass-count replay decision agrees with itself and $c_H(x_N, N) = 0$. Terminal stops add zero replay flip for every true count; this endpoint convention is shared by Theorem 1 and the DP in Appendix B.

9 CONCLUSION

Treating across-task heterogeneity as the certificate object turns early stopping for majority voting into an exact conditional-inference problem under the replay model: adaptivity is free (tower property), calibration is one empirical-Bernstein step over problems, and the supplied OMR readout has an 81% replay rollout-count reduction at full-vote flip level 0.05, where the strongest distribution-free baseline has 16% and the strongest fixed mixture-aware budget 47%. The same framework admits a synthetic ordered-drift analysis with a margin repair, while natural online order, gold correctness, and actual token/latency/cancellation costs remain open.

REPRODUCIBILITY STATEMENT

The count-replay analyses, assumptions, split protocol, and complete proofs are specified in Sections 3–5 and the appendix. The supplied artifacts, pinned carrier checksums, deterministic seeds, and claim checker are indexed in the anonymous supplementary reproduction package; its current limitations, including unavailable upstream carrier paths and the absence of ordered online telemetry, are stated in Section 8.

AI-USE STATEMENT

Generative AI tools were used to assist manuscript-language revision, evidence and citation auditing, build checks, and internal pre-submission review, including the Codex multi-agent revision and audit documented for this version. A generative image tool was also used to compose Figure 1's conceptual layout; its quantitative labels were independently checked against the cited on-disk result artifacts. These tools did not generate, execute, or replace the reported scientific experiments, their quantitative result artifacts or values, or the final technical judgments; the authors reviewed the changes and remain responsible for the paper.

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A Proofs

A.1 Monotonicity of the fixed-k flip probability (Lemma 1)

Recall

$$f_K(k) = \mathbb{P}\big(\mathrm{MV}(k) \neq \mathrm{MV}(N) \,\big|\, K\big), \qquad x_k \sim \mathrm{HG}(N, K, k),$$

with the negative side winning ties ($\mathrm{MV}(k) = 1$ iff $x_k > k/2$). We prove that $f_K(k)$ is nonincreasing along the odd numbers $k \in \{1, 3, 5, \ldots, N-1\}$ for every fixed K; this suffices for the paper, since the fixed-k certificate family is evaluated on odd budgets only, and it makes $k^*(\alpha) = \min\{k : \mathbb{E}_H[f_K(k)] \leq \alpha\}$ well defined and binary-searchable for every mixture H (mixtures of nonincreasing functions are nonincreasing).

Setup and case split. $\mathrm{MV}(k) \neq \mathrm{MV}(N)$ can happen in only two ways:

- **Case U (upset).** $K \leq N/2$ (full vote negative) and $x_k > k/2$ (prefix votes positive).
- **Case R (reversal).** $K > N/2$ (full vote positive) and $x_k \leq k/2$ (prefix votes negative).

Write $f_K(k) = U_K(k) + R_K(k)$ with $U_K(k) = \mathbb{P}(x_k > k/2)\,\mathbf{1}\{K \leq N/2\}$ and $R_K(k) = \mathbb{P}(x_k \leq k/2)\,\mathbf{1}\{K > N/2\}$. At most one of the two terms is nonzero for any K.

The reflection identity. Swapping the labels "pass"/"fail" of all N rollouts maps the law $\mathrm{HG}(N, K, k)$ of x_k onto the law of $k - x_k$ under count $N - K$, i.e. $\mathbb{P}_K(x_k > s) = \mathbb{P}_{N-K}(x_k < k - s)$. Hence, for odd k,

$$U_K(k) = \mathbb{P}_K(x_k > k/2) = \mathbb{P}_{N-K}(x_k < k/2) = R_{N-K}(k) \qquad (K \leq N/2),$$

since k odd makes $k/2$ a half-integer, so the strict and weak inequalities coincide. Every upset probability thus equals a reversal probability at the reflected count, and it suffices to prove

$$R_K(k+2) \leq R_K(k) \qquad \text{for all } K > N/2 \text{ and odd } k, \tag{3}$$

which covers Case R directly and Case U by reflection (the same display applied at count $N - K$).

Coupling construction. Fix $K > N/2$ and odd $k \leq N - 3$. We couple the two hypergeometric draws on a single probability space. Let the population of N rollouts consist of K passes and $N - K$ fails. Draw an *ordered* sample of size $k + 2$ uniformly without replacement and split it as

$$\underbrace{X_1, \ldots, X_k}_{\text{prefix } k} \;\Big|\; \underbrace{X_{k+1}, X_{k+2}}_{\text{extra pair } (A,B)}.$$

The marginal law of $(X_1, \ldots, X_k)$ is exactly $\mathrm{HG}(N, K, k)$, and $x_{k+2} = x_k + A + B$ has law $\mathrm{HG}(N, K, k+2)$. All statements below are about this joint space, so they hold for the two marginals simultaneously.

Let $t = k/2 + 1/2$ be the (integer) flip threshold: the prefix at k votes negative iff $x_k \leq t - 1$; the prefix at $k + 2$ votes negative iff $x_{k+2} \leq t$. Partition the event $\{x_k = s\}$ by the value of the pair $A + B \in \{0, 1, 2\}$:

$$\begin{aligned}
E_0 &= \{x_k = t-1,\ A+B = 0\}, && \text{flip at } k,\ \text{flip at } k+2,\\
E_1 &= \{x_k = t-1,\ A+B = 1\}, && \text{flip at } k,\ \text{flip at } k+2,\\
E_2 &= \{x_k = t-1,\ A+B = 2\}, && \text{flip at } k,\ \text{no flip at } k+2,\\
F_0 &= \{x_k = t,\quad A+B = 0\}, && \text{no flip at } k,\ \text{flip at } k+2,\\
F_1 &= \{x_k = t,\quad A+B = 1\}, && \text{no flip at } k,\ \text{no flip at } k+2.
\end{aligned}$$

All other states agree on both sides (e.g. $x_k \leq t-2$ flips under both, since $A + B \leq 2$; $x_k \geq t + 1$ flips under neither). Therefore

$$R_K(k) - R_K(k+2) = \mathbb{P}(E_2) - \mathbb{P}(F_0), \tag{4}$$

and it remains to show $\mathbb{P}(E_2) \geq \mathbb{P}(F_0)$.

Comparison of the two boundary atoms, including support boundaries. Both atoms factor as (prefix probability) $\times$ (pair probability). We first split off the case $\mathbb{P}(F_0) = 0$. Then (4) gives $R_K(k) - R_K(k+2) = \mathbb{P}(E_2) \geq 0$ directly; in particular, no ratio is formed when a binomial factor in F_0 vanishes (including a zero remaining-fail count or an infeasible $x_k = t$ atom). It remains to consider $\mathbb{P}(F_0) > 0$. This support condition makes every denominator below strictly positive and makes the ratio algebra legitimate. With $k = 2t - 1$, first,

$$\mathbb{P}(x_k = t-1) = \frac{\binom{K}{t-1}\binom{N-K}{t}}{\binom{N}{k}}, \qquad \mathbb{P}(x_k = t) = \frac{\binom{K}{t}\binom{N-K}{t-1}}{\binom{N}{k}},$$

since $k - (t-1) = t$ and $k - t = t - 1$. Given $x_k = t - 1$, the remainder holds $K - t + 1$ passes out of $N - k$, so $\mathbb{P}(A + B = 2 \mid x_k = t-1) = \binom{K-t+1}{2}/\binom{N-k}{2}$; given $x_k = t$, the remainder holds $N - K - t + 1$ fails, so $\mathbb{P}(A + B = 0 \mid x_k = t) = \binom{N-K-t+1}{2}/\binom{N-k}{2}$. The denominators $\binom{N}{k}\binom{N-k}{2}$ are common to both atoms. Under $\mathbb{P}(F_0) > 0$, hence

$$\frac{\mathbb{P}(E_2)}{\mathbb{P}(F_0)} = \frac{\binom{K}{t-1}\binom{N-K}{t}\binom{K-t+1}{2}}{\binom{K}{t}\binom{N-K}{t-1}\binom{N-K-t+1}{2}} = \frac{K-t}{N-K-t},$$

where the last equality uses the three elementary ratios $\binom{K}{t-1}/\binom{K}{t} = \frac{t}{K-t+1}$, $\binom{N-K}{t}/\binom{N-K}{t-1} = \frac{N-K-t+1}{t}$, and $\binom{K-t+1}{2}/\binom{N-K-t+1}{2} = \frac{(K-t+1)(K-t)}{(N-K-t+1)(N-K-t)}$, whose product telescopes to the displayed ratio. Therefore, with $C := \binom{K}{t}\binom{N-K}{t-1}\binom{N-K-t+1}{2}/\big[\binom{N}{k}\binom{N-k}{2}\big] \geq 0$,

$$\mathbb{P}(E_2) - \mathbb{P}(F_0) = C\big[(K-t) - (N-K-t)\big] = C\,(2K - N) \;\geq\; 0, \tag{5}$$

because $K > N/2$ in Case R. The excluded support case was already proved directly; if $t > K$, $\mathbb{P}(x_k = t-1) = 0$ and both atoms vanish. By (4) this proves (3) and, by the reflection identity, $U_K(k+2) \leq U_K(k)$ as well, completing the proof of monotonicity along odd k. ☐

Remark 4 (Verification at $N = 32$). Independently of the proof, the revision's exact-rational CPU audit enumerates all admissible (N, K, k) for $2 \leq N \leq 64$, including the boundary $(N, K, k) = (32, 17, 29)$, checks (4), the support split, and the monotonicity inequality. The historical $N = 32$ artifact remains a separate frozen result check.

Remark 5 (Why only odd k enters the certificate). Requiring k odd removes tie-breaking from the certificate family (every prefix has a strict majority side), and the binary search for $k^*(\alpha)$ is over the odd ladder only. Nothing else in the paper uses monotonicity at even k.

A.2 TERMINAL-STOP BOOKKEEPING (THEOREM 1)

The stopping rule is $\tau = \min\{k \in \{3, \ldots, N\} : c_H(x_k, k) \leq \alpha\}$. At a reachable terminal state, all N binary outcomes have been observed, hence $x_N = K$. We define the endpoint certificate to be zero; whenever the posterior is supported, this also follows because the hypergeometric likelihood at (x_N, N) has support only on $K = x_N$:

$$c_H(x_N, N) = 0.$$

Thus $k = N$ satisfies the same threshold rule rather than being an exceptional terminal state. The tower argument applies to this bounded stopping time, and

$$\mathbb{P}(\text{flip}) = \mathbb{E}\big[c_H(x_\tau, \tau)\big] = \mathbb{E}\big[c_H\, \mathbf{1}\{\tau < N\}\big] + \mathbb{E}\big[c_H\, \mathbf{1}\{\tau = N\}\big].$$

The first term is at most $\alpha\,\mathbb{P}(\tau < N)$, and the second term is zero. Therefore the integrand is bounded pointwise by α and $\mathbb{P}(\text{flip}) \leq \alpha$. For each fixed true count K, a path reaching N also has $x_N = K$, so its terminal contribution to the DP flip probability $g(K; \alpha, \widehat{H}_m)$ is exactly zero. The population layer remains needed for calibration-prior estimation, not to absorb a terminal flip.

Remark 6 (The terminal-stop mass is small in practice). On the TEST split, the exact fraction of paths reaching $k = N$ under BAYES-H is 0.59% at $\alpha = 0.05$ and 1.10% at $\alpha = 0.02$ (the certificate stops early exactly on the bimodal mass). Their contribution to replay flip is zero; their contribution to the expected stopping count is included exactly (Appendix B).

A.3 LINEARITY OF THE POPULATION FLIP RATE

For fixed rule and fixed prior $\widehat{H}_m$ used *inside* the certificate, the per-task flip probability $g(K)$ depends only on the true count K (via the hypergeometric law of prefixes) and on $\widehat{H}_m$ (through the posterior threshold), but is a fixed function once $\widehat{H}_m$ is frozen after FIT. The population replay flip rate is therefore

$$\mathbb{P}_{\text{replay},H}(\text{flip}) = \sum_{K=0}^{N} H[K]\, g(K),$$

a *linear* functional of the true mixture H. Consequences used in the main text: (i) no Jensen gap — bounding $\bar{g}_{\text{CAL}}$ by empirical Bernstein bounds the population quantity directly, with estimation error of $\widehat{H}_m$ entering only through the empirical mean of a bounded per-task quantity $g \in [0,1]$; (ii) graceful degradation — if the deployment mixture shifts from H to H', $|\mathbb{P}_{H'} - \mathbb{P}_H| \le \|H' - H\|_1$ since $g \in [0,1]$; (iii) Bonferroni over the pre-declared family of $J = 64$ rules legitimizes the min-k selection, because simultaneous UCBs on all J linear functionals imply the UCB of the selected rule.

B EXACT DYNAMIC PROGRAM FOR THE ADAPTIVE RULE

We compute, for a fixed true count K and frozen prior $\widehat{H} = \widehat{H}_m$, the exact per-task flip probability $g(K; \alpha, \widehat{H})$ and expected stopping time $\mathbb{E}[\tau \mid K]$ of BAYES-H under the replay model, by dynamic programming over prefix states.

State space and transitions. A state is a prefix (k, x) with $0 \le x \le k \le N$, $k \ge 0$. Under count K, the next draw is a uniform element of the remainder, so the transition kernel is

$$\mathbb{P}(x_{k+1} = x + 1 \mid k, x, K) = \frac{K - x}{N - k}, \qquad \mathbb{P}(x_{k+1} = x \mid k, x, K) = \frac{N - K - (k - x)}{N - k},$$

and the initial state has $\mathbb{P}(x_0 = 0) = 1$. The state graph is a DAG ($O(N^2)$ states, $O(N^2)$ edges), so a single backward pass is linear in the number of states.

Certificate table. The posterior $\mathbb{P}(K = j \mid k, x) \propto \widehat{H}[j]\,\mathrm{HG}(N, j, k, x)$ and $c_H(x, k)$ are pre-computed once for all (k, x) on the $O(N^2)$ grid (with the terminal convention $c_H(x, N) := 0$). At a posterior-supported endpoint, the likelihood supports only $j = x$, so this convention is the conditional flip probability; it also defines a zero-prior-mass table row, where observing all outcomes still fixes the full-N binary pass-count decision. A state is a *stop state* iff $k \ge 3$ and $c_H(x, k) \le \alpha$; therefore $k = N$ satisfies the same stopping test, rather than a separate truncation rule.

Recursions. For a fixed K, define $V(k, x) = \mathbb{P}(\mathrm{MV}(\tau) \ne \mathrm{MV}(N) \mid k, x, K)$ and $T(k, x) = \mathbb{E}[\tau - k \mid k, x, K]$. Then

$$V(k,x) = \begin{cases} \mathbf{1}\{\mathrm{side}(x,k) \ne \mathrm{side}(K,N)\}, & (k,x) \text{ stop state}, \\ \dfrac{K - x}{N - k}\, V(k+1, x+1) + \dfrac{N - K - k + x}{N - k}\, V(k+1, x), & \text{otherwise}, \end{cases}$$

and the same recursion with $T(\cdot) = 0$ at stop states and $T(k, x) = 1 +$ (continuation mean) otherwise. The requested quantities are $g(K) = V(0, 0)$ and $\mathbb{E}[\tau \mid K] = T(0, 0)$. The per-task exact saving is $1 - \mathbb{E}[\tau \mid K]/N$; population quantities follow by linearity, $\sum_K H[K] g(K)$, for any mixture H. The same forward pass yields the exact terminal-stop occupancy $\mathbb{P}(\tau = N \mid K)$. At that boundary the reachable state has $x = K$, so the stop-state value in the recursion is zero for replay flip; on the TEST split terminal occupancy is 0.59% at $\alpha = 0.05$ and 1.10% at $\alpha = 0.02$.

Complexity. For each K: $O(N^2)$ states, $O(1)$ work per state after the certificate table is frozen, so all 33 values of K cost $O(N^3) \approx 3.6 \times 10^4$ flops at $N = 32$ — negligible. This is why every TEST number in the main table is an *exact replay quantity*, not a Monte Carlo estimate.

Cross-check against simulation. The DP is validated against an independent replay Monte Carlo implementation (fresh random prefixes) on the same frozen prior: the maximum absolute deviation over the checked cells is 0.043, within the pre-set tolerance $3\sigma_{\mathrm{MC}} = 0.075$ (the check includes cells at $\alpha = 0.01$, whose realized flips are small and inflate the pooled MC standard error); PASS. See `fit_cal_test_r469.py` and the self-check block of `fit_cal_test_r469_result.json`.

C FEATURE COMPARISON AND APPENDIX ROADMAP

This appendix separates proof (Appendix A), exact DP bookkeeping (Appendix B), synthetic diagnostics, and fixed-rule sensitivity. None expands the main count-exchangeable replay claim to online correctness or operational cost.

Work	Endpoint	Order/model	Guarantee / selection
ASC (Aggarwal et al., 2023); ESC (Li et al., 2024)	adaptive self-consistency	implementation-specific sequential sampling	no claim here about their universal full-vote certificate
MARS (Chen et al., 2026)	parallel trace vote movement	parallel traces/checkpoints	conservative future-movement bound; distinct from this replay posterior
Conformal (Vovk et al., 2005); CS (Howard et al., 2021)	coverage / time-uniform mean bounds	exchangeable / i.i.d.-sequential	established marginal or anytime tools, not this conditional flip object
This work	full-N binary pass-count replay disagreement	count-exchangeable random prefix	posterior flip certificate; FIT/CAL rule selection, TEST replay readout

Table 2: Source-verified feature comparison. Relationships are deliberately endpoint- and model-specific; this table makes no priority or ordered-online claim.

D FULL DRIFT TABLES

Tables 3 and 4 report the complete drift stress grid behind Section 7 (`drift_stress_r469_result.json`; 2000 TEST problems, 500 orders per cell). Under the smooth mechanisms E2/E3, BAYES-H is the most robust stopper at both levels; under the premise-breaking E1 front-load all certificates fail together, as they should.

Table 3: Full drift stress table at $\alpha = 0.05$ (2000 TEST problems, 500 orders each; flip in %, mean budget k in parentheses). **Bold**: realized flip exceeds α (certificate broken).

Mechanism	δ	BAYES-H	FIXED-EB	FIXED-HOEF	WINDOW3
E1 (front-load)	0.0	2.5 (6.1)	3.2 (17.0)	1.6 (27.0)	**5.8 (4.4)**
	0.05	4.8 (6.1)	4.9 (17.0)	4.1 (27.0)	**7.1 (4.4)**
	0.1	**7.6 (6.2)**	**8.2 (17.0)**	**7.6 (27.0)**	**9.6 (4.3)**
	0.15	**11.4 (6.0)**	**12.1 (17.0)**	**12.4 (27.0)**	**12.2 (4.2)**
	0.2	**13.0 (5.5)**	**13.7 (17.0)**	**13.6 (27.0)**	**13.5 (4.1)**
E2 (linear trend)	0.0	4.7 (6.1)	4.7 (17.0)	3.8 (27.0)	**6.8 (4.4)**
	0.05	4.8 (6.2)	4.7 (17.0)	3.7 (27.0)	**7.0 (4.4)**
	0.1	**5.2 (6.3)**	4.9 (17.0)	3.7 (27.0)	**7.6 (4.5)**
	0.15	**5.9 (6.4)**	**5.3 (17.0)**	3.7 (27.0)	**8.8 (4.6)**
	0.2	**7.0 (6.5)**	**6.0 (17.0)**	3.7 (27.0)	**10.4 (4.6)**
E3 (adversarial block)	0.0	4.7 (6.1)	4.7 (17.0)	3.8 (27.0)	**6.8 (4.4)**
	0.05	4.8 (6.1)	4.7 (17.0)	3.8 (27.0)	**6.8 (4.4)**
	0.1	4.9 (6.1)	4.7 (17.0)	3.8 (27.0)	**6.8 (4.4)**
	0.15	**5.2 (6.1)**	4.7 (17.0)	3.8 (27.0)	**6.8 (4.4)**
	0.2	**5.5 (6.1)**	4.7 (17.0)	4.2 (27.0)	**6.8 (4.4)**

Table 4: Full drift stress table at $\alpha = 0.1$ (2000 TEST problems, 500 orders each; flip in %, mean budget k in parentheses). **Bold**: realized flip exceeds α (certificate broken).

Mechanism	δ	BAYES-H	FIXED-EB	FIXED-HOEF	WINDOW3
E1 (front-load)	0.0	3.7 (5.1)	7.9 (5.0)	6.4 (7.0)	5.8 (4.4)
	0.05	5.6 (5.1)	9.1 (5.0)	7.8 (7.0)	7.1 (4.4)
	0.1	8.4 (5.2)	**11.0 (5.0)**	**10.1 (7.0)**	9.6 (4.3)
	0.15	**11.9 (5.0)**	**12.7 (5.0)**	**12.3 (7.0)**	**12.2 (4.2)**
	0.2	**13.3 (4.7)**	**13.7 (5.0)**	**13.5 (7.0)**	**13.5 (4.1)**
E2 (linear trend)	0.0	5.4 (5.1)	8.7 (5.0)	7.3 (7.0)	6.8 (4.4)
	0.05	5.5 (5.2)	8.9 (5.0)	7.5 (7.0)	7.0 (4.4)
	0.1	6.0 (5.3)	10.0 (5.0)	8.3 (7.0)	7.6 (4.5)
	0.15	6.9 (5.3)	**11.6 (5.0)**	9.7 (7.0)	8.8 (4.6)
	0.2	8.3 (5.4)	**13.7 (5.0)**	**11.6 (7.0)**	**10.4 (4.6)**
E3 (adversarial block)	0.0	5.4 (5.1)	8.7 (5.0)	7.3 (7.0)	6.8 (4.4)
	0.05	5.4 (5.1)	8.6 (5.0)	7.3 (7.0)	6.8 (4.4)
	0.1	5.5 (5.1)	8.7 (5.0)	7.3 (7.0)	6.8 (4.4)
	0.15	5.6 (5.1)	8.7 (5.0)	7.4 (7.0)	6.8 (4.4)
	0.2	5.7 (5.1)	8.7 (5.0)	7.3 (7.0)	6.8 (4.4)

E THE CRITICAL-RADIUS CONSERVATION CURVE

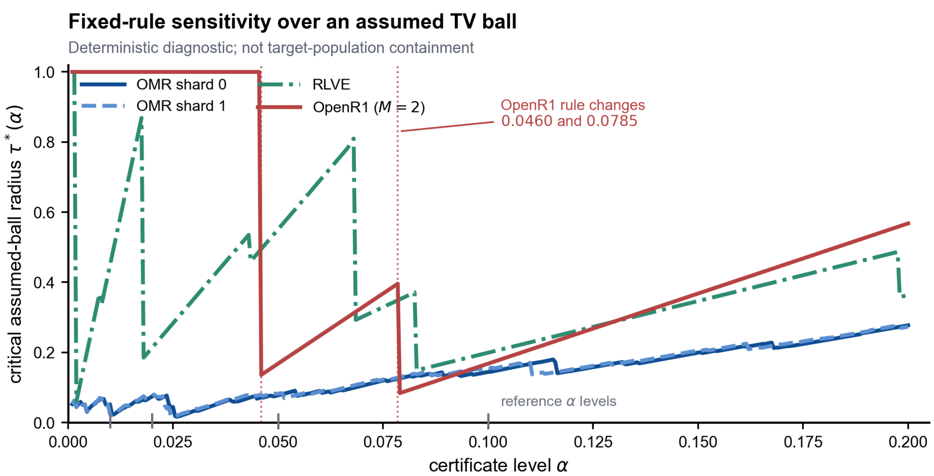

Figure 2: Deterministic fixed-rule sensitivity radius $\tau^*(\alpha)$ over an assumed fitted-mixture TV ball; this is not a confidence set for, or a containment claim about, an unknown target population. Proposition 1 is evaluated on the continuum $\alpha \in [10^{-3}, 0.2]$ (step 5×10^{-4}), computed exactly per level from the frozen g-profiles of the three carriers (OMR both FIT shards; OpenR1 exact in closed form). Grey tick marks: the four reference levels; red dotted lines: the OpenR1 certificate thresholds $0.0785/0.0460$ where the frozen rule changes. Artifact: `tv_conservation_r484_result.json`.

Figure 2 makes the rule-change structure of the certificate visible. *(a) Staircase law.* Within any α-interval on which the frozen rule is unchanged, the whole vector g — hence $V(R)$ — is constant, so $\tau^*(\alpha) = \max\{R : V(R) \leq \alpha\}$ is *non-decreasing*; every strict decrease of τ^* coincides with a rule-change breakpoint (verified at every grid cell of every carrier; the rule-change counts on the plotted range are 96/93/7/2 for OMR shard 0 / OMR shard 1 / RLVE / OpenR1). *(b) Analytic breakpoints on OpenR1.* With $V(R) = \mathrm{flip}(\widehat{H}) + g_{\max} R$ exactly, $\tau^*(\alpha) = \alpha/g_{\max} - \widehat{H}(\frac{1}{2})$: at $\alpha = 0.07852$ the fail-prefix stop engages and $g_{\max}$ doubles ($0.125 \to 0.25$), so τ^* jumps *down* $0.396 \to 0.082$ ($\Delta = -0.314$); at $\alpha = 0.04598$ the pass-prefix stop disengages and the frozen rule degenerates to never-stop ($V(R) \equiv 0$ for any prior), so the whole-simplex plateau $\tau^* = 1$

holds *below* this threshold and τ^* jumps *down* $1 \to 0.136$ ($\Delta = -0.864$) as α crosses it upward. *(c) Whole-simplex plateaus.* Wherever the frozen rule's worst case cannot exceed α for *any* prior ($\max_K g(K) \le \alpha$), $\tau^* = 1$: RLVE for $\alpha < 0.002$ and OpenR1 for $\alpha < 0.0460$; OMR's rule never stops only below 10^{-3}, off the plotted range. *(d) Alignment co-variation (decomposed).* The zero-g alignment number co-varies with τ^* at pooled Spearman 0.75 across the 16 carrier$\times$level cells (descriptive; the per-cell alignment is itself α-dependent). The decomposition audit (`spearman_decomposition_r486_result.json`) shows this is a between-carrier law, not a within-carrier one: 1.0 across the four carrier means, 0.63 on the 14 non-plateau cells, 0.44–0.90 under leave-one-carrier-out; within-carrier it is -0.95/-0.89 (OMR shards), $+0.32$ (RLVE), $+1.0$ (OpenR1) — the OMR reversal is the staircase mechanism itself (level tightening adds zero-g mass on the lowest-g atoms only). Heterogeneity is not a confounder: het-excess vs τ^* Spearman 0.01, het-excess vs alignment 0.05 (16 cells). *Practical reading.* The staircase shape is a design signal for choosing the certificate level: inside a constant-rule interval, loosening α buys certified drift tolerance gradually; at a breakpoint it buys (or loses) a discrete jump — α should be set on a step, not inside one. *(e) How much FIT data buys how much robustness.* Rebuilding the prior from nested prefixes of the frozen FIT order at $m \in \{\frac{1}{32}, \ldots, 1\} \times m_{\text{full}}$ and recomputing $\tau^*_m(\alpha)$ exactly (`prior_fit_size_r491_result.json`) gives an operational size rule. On OMR the $\tau^*_m(0.05)$ curves already clear the observed cross-shard transfer 0.042 at the smallest tested pool m=125 (shard 0: 0.062; shard 1: 0.046), i.e. cross-shard robustness does *not* require the full 4000-problem fit. The radius is *not* monotone in m (OMR shard 1 at $\alpha = 0.01$: 0.010 at m=125, up to 0.056 at m=1000, back to 0.024 at m=4000): growing the fit moves both the prior (better) and the frozen rule (larger $g_{\max}$ possible), and the rule channel can dominate — τ^*_m is a property of the fitted *pair* $(\widehat{H}_m, \text{rule}_m)$, not of the sample size alone. The closed loop of Proposition 1 holds at every one of the 51 cells with $\text{TV}(\widehat{H}_m, \widehat{H}_{\text{full}}) \le \tau^*_m$: the m-rule's exact flip under the full-data prior never exceeds α (0 violations), while at the coarsest OMR shard 1 fit (m=125, TV 0.175, outside the ball) the transferred rule genuinely breaks (flip $0.0244 > 0.02$, $0.0165 > 0.01$) and is repaired from m=250 onward — inside/outside the radius predicts validity/failure exactly. The estimation error itself scales as $\text{TV}_m \approx C/\sqrt{m}$ (within-carrier spread of $\text{TV}_m\sqrt{m}$ below 1.5$\times$ on OMR; the RLVE spread reaches 3.5$\times$ only because its coarsest prefixes carry a missing-mode error $\sim\sqrt{m_{\text{full}}/m}$ larger). *(f) Which channel moves τ^*_m: rule, not prior — with one caveat.* We decompose the m-sensitivity into its two moving parts by computing, at every (carrier, m, α) cell, the two counterfactual radii (`rule_channel_r494_result.json`): $\tau^{*\,\text{pf}}_m = \tau^*(\widehat{H}_{\text{full}}, g_m)$ freezes the prior at the full fit and lets only the *rule* vary with m, and $\tau^{*\,\text{rf}}_m = \tau^*(\widehat{H}_m, g_{\text{full}})$ freezes the rule and lets only the *prior* vary. The recomputed both-moving curve reproduces the m-grid of `prior_fit_size_r491_result.json` bit-exactly (0/72 mismatches). Three findings. *(i) The rule channel drives the non-monotonicity.* With the prior frozen at $\widehat{H}_{\text{full}}$, $\tau^{*\,\text{pf}}_m$ is still non-monotone at OMR shard 1 α=0.01: 0.000 at m=125 rising to 0.060 at m=1000 then falling to 0.024 at m=4000 — the same up-then-down shape as the both-moving curve — because the refitted rule's $g_{\max}$ itself varies non-monotonically ($0.563 \to 0.075 \to 0.098$) as the coarse prefix shifts which budgets are certifiable. *(ii) The estimation channel is benign on average but* not *uniformly.* Holding the rule fixed, a coarser prior can *shrink* the radius below the full-fit value in 31 of 72 cells (all on OMR shard 1 and RLVE; largest deficit 0.053 at RLVE m=93, α=0.05). The mechanism is the base-flip term $\sum_K \widehat{H}[K]\, g(K)$: a coarse prefix that under-covers the likely-correct (low-g) region and over-weights high-g mass raises the base flip, and — because the *frozen full rule* already has large g exactly on those atoms — the certificate has less slack to spend on drift. This is the missing-mode failure of (e) seen through the rule: the same coarse prefix that carries a $\sim\sqrt{m_{\text{full}}/m}$ estimation error also pays it on the worst (largest-g) coordinates. *(iii) The RLVE coarsest dip is pure rule-channel.* The collapse of the both-moving radius at RLVE m=187, α=0.05 (0.235, far below both m=93 at 0.616 and m=375 at 0.534) survives with the prior frozen ($\tau^{*\,\text{pf}}_{187} = 0.222$) and vanishes with the rule frozen ($\tau^{*\,\text{rf}}_{187} = 0.530$): it is entirely a $g_{\max}$ event (0.179 vs. 0.071 for the full rule), not an estimation one. *Reading.* The robustness certificate is governed by the *rule* the fit induces more than by the prior's sampling error: two fits of the same size can certify very different radii because the induced rule differs, and the only lever that reliably buys radius is one that shrinks the induced g-profile (e.g. the margin repair of Section 7), not one that merely adds FIT data. *(g) A constructive repair that buys radius: flip-budget state subsets.* The reading of (f) — shrink the induced g-profile — is here made executable (`full_sweep_r498e_result.json`). Fix a per-atom flip budget c. On each atom K, order the original rule's stopping states by their certificate value and keep the longest prefix whose cumulative flip contribution is at most c; states not kept run to n. The realized subset rule satisfies, pointwise on every atom, $g_S(K) \le \min(g_{\rm orig}(K), c)$ (domination), its stopping set is a subset of the original rule's (it never stops earlier, so the replay-count cost is only extra votes), and its exact radius is recomputed by the same linear program — nothing is re-estimated from data. Sweeping $c \in \{0.30, \ldots, 0.01\}$ over all 72 carrier$\times m \times \alpha$ cells of (e): recomputation anchors the grid bit-exactly (0/72 mismatches), domination holds at every atom of every cell, and at the best cap every cell reaches $\tau^* = 1$: for 70 of 72 cells this is a strict gain over the original radius at $+1\%$ to $+11\%$ mean votes (max $+11.3\%$ at OMR shard 0 m=500, α=0.10; RLVE never exceeds $+3.3\%$), while the remaining two (RLVE at α=0.02) are already whole-simplex under the original rule and the sweep selects no cap. The $\tau^* = 1$-by-cap cells sit in the *trivial-validity* regime: the capped profile satisfies $\max_K g_S(K) \le \alpha$ *and* $\mathrm{base}_S \le \alpha$ by construction, so the certificate survives *any* bounded prior shift. This is not "infinite robustness for free": the bill is paid by the realized base-flip level base_S (e.g. $0.0054 \ll \alpha$=0.05 at OMR m=4000, cap 0.02) and by the extra votes, both reported per cell. The pitfall cell of (f) is repaired exactly: RLVE m=187, α=0.05 goes from $\tau^* = 0.235$ to 1.0 at cap 0.07 with $+0.5\%$ votes; the coarse OMR shard 1 m=125 fits all exceed half their full-fit radius at every level (e.g. α=0.01: $0.0095 \to 1.0$). Two negative results delimit the mechanism. First, *support truncation fails*: zeroing the low-mass atoms of the prior and renormalizing changes the induced certificate not at all (bit-identical g-profiles, 0 differences in 6 probed cells $\times$ 5 truncation levels; `support_trunc_repair_r498_result.json`) — the spikes of (f) live on near-zero-mass boundary atoms that the base term never visits but the ℓ_1 adversary targets, and removing them from the prior does not touch the local likelihood geometry that induces them. Second, *deadline truncation fails*: capping each atom's stopping time at $\kappa(K)$ is non-monotone in κ (flip 0.014 at κ=8 rising to 0.357 at κ=7 and 0.5 at κ=5 on a probed RLVE atom) and can certify *less* than the original rule (RLVE m=187, α=0.05: $0.235 \to 0.133$; `efficiency_r498c_result.json`) — a hard deadline deletes informative late states wholesale, whereas the certificate-order subset deletes exactly the states that are expensive to certify. Together with (f) this closes the loop: moving the prior does nothing, moving the time axis hurts, moving the *stopping-state set* in certificate order repairs — the rule channel is the lever, and the effective control is a flip budget, not a data budget. The repair is also not a degenerate retreat to stopping at the first informative prefix: a fixed k=3 stopper on OMR shard 1 m=125 has base flip 0.093 ($9\times$ above α=0.01, $\tau^* = 0$) against the repaired rule's $\mathrm{base}_S = 0.0036$. *Universal budget.* The sweep also pins down a single carrier- and m-independent repair budget (`universal_cap_r500_result.json`): at cap 0.01 — the smallest level of the paper's grid — *every* one of the 72 cells reaches $\tau^* = 1$, at $+1.0\%$ to $+22.3\%$ mean votes (OMR; RLVE $\le +5.2\%$) and realized base flip ≤ 0.0038. The boundary is sharp within the scanned grid: at cap 0.015 exactly the 18 cells with α=0.01 (6 per carrier, every m) still fall short of the whole simplex, while all 54 cells with $\alpha \ge 0.02$ are already repaired. A budget at or below the smallest level of interest therefore repairs every fit size and every carrier we tested, and no smaller scanned budget suffices for the strictest level — the repair law is a property of the cap relative to α, not of m or the carrier. A fine-scale pass (`critical_cap_r503_result.json`; caps at 5×10^{-4} resolution plus bisection to $\approx 6 \times 10^{-7}$ on the full-radius prefix) sharpens this into an exact edge law. Because domination gives $\max_K g_S(K) \le c$ pointwise, every cap $c \le \alpha$ certifies the whole simplex mechanically, so the effective critical cap $c^* = \sup\{c : \tau^*(c) = 1\}$ satisfies $c^* \ge \alpha$ in every cell; the supremum is attained strictly above α whenever the realized staircase leaves $\max_K g_S(K) \le \alpha$ beyond $c = \alpha$ — which happens in all 72 cells. The margin above α is carrier-resolved: it stays $\le +0.8\%$ at $\alpha \ge 0.02$ on OMR (one shard 1 cell excepted, $+2.8\%$) and reaches $+10.6\%$ at α=0.01 on OMR (fine N=32 certificate geometry). On RLVE the margin is not just measured but *derived* (`edge_law_r504_result.json`): the flip mass of stopping at state (k, x) on an atom is the hypergeometric reach probability $\binom{K}{x}\binom{n-K}{k-x}/\binom{n}{k}$, a rational number that does not depend on the fitted prior; the smallest certificate-greedy achievable flip sum above α is thus computable from the certificate table alone, and it equals the scan-measured edge wherever the scan bracketed one (all 18 such cells). On RLVE (N=8) this quantum sits at c^*=3/28 at α=0.10 ($15/14 \cdot \alpha$), $1/14$ at α=0.05 ($10/7 \cdot \alpha$), and $1/70$ at α=0.01 ($10/7 \cdot \alpha$), identical at every one of the six fit sizes — the edge is combinatorics of the binding atom, not of the prior. At α=0.02 the cell splits by a certificate straddle: at the four fit sizes whose binding state $(4, 1)$ has certificate $\le$0.02 the edge is $1/14$ ($25/7 \cdot \alpha$, beyond the scanned band, which had reported only the censored bound $c^* \ge 2\alpha$), while at $m \in \{93, 750\}$ that state's certificate crosses α=0.02 under the coarse-prefix prior, every achievable flip sum stays $\le \alpha$, and the full radius survives *every* budget (τ^*=1 certified at cap 0.5, $\max_K g_S$=1/56). On OMR (N=32) no such $O(\alpha)$ gap exists: the per-atom closed-form edge upper-bounds the budget-constrained scan edge (equality at 11 of 48 cells) and stays within $+10.6\%$ of α in every cell — the quantization of the repair edge is a small-N phenomenon. A tightness audit of the bound (`edge_tightness_r505_result.json`) separates the 48 OMR cells into two regimes by the sign of the certified slack one step below the closed edge: in 11 cells the realized $\max_K g_S$ at cap $c^*_{\text{closed}} - \epsilon$ is still $\leq \alpha$ (the binding atom's crossing state has not yet entered), so the scan edge and the closed edge coincide up to the scan's bracket resolution — the closed form is tight; in the remaining 37 cells the realized $\max_K g_S$ at $c^*_{\text{closed}} - \epsilon$ already exceeds α by 2.9×10^{-5} to 3.7×10^{-3}, so the budget-constrained greedy must drop the crossing prefix and $c^*_{\text{scan}} < c^*_{\text{closed}}$ strictly. The two regimes are disjoint in sign (nearest slacks -2.43×10^{-5} and $+2.91 \times 10^{-5}$, an inner margin ratio of $1.20\times$ — correcting the $\geq 2.4\times$ printed in v5.19–v5.20), so tightness is decided by a sign, not a fitted threshold. (This audit also corrected the equality count reported above in v5.18 from 12 to 11: the twelfth cell, shard 1 $m{=}2000$ $\alpha{=}0.10$, differs by 8.2×10^{-5} — two orders of magnitude above the scan's rounding noise — and had been absorbed by the verifier's 10^{-4} tolerance.) A discriminating audit at sub-artifact resolution (`discriminant_r507_result.json`) head-to-heads the slack-sign classifier against the two falsified first-draft conditions (recorded in `edge_tightness_r505_result.json`) on all 48 cells: the sign rule is exact (11 TP / 37 TN, no errors, invariant to the probe offset over $[5 \times 10^{-8}, 10^{-6}]$); the unique-binding-atom biconditional errs on 22 cells (19 FP $+3$ FN); and the best fixed margin threshold achieves only the always-strict constant (11 errors). The audit also maps the tolerance window itself: the 11-cell tight count is stable on $[5 \times 10^{-7}, 2 \times 10^{-6}]$, degenerates to 3 at 2.5×10^{-7} (the scan edge's six-decimal storage spreads exact equalities to $\pm 4 \times 10^{-7}$), and would absorb the genuine twelfth cell only above 8.22×10^{-5} — the usable discriminating band is $[5 \times 10^{-7}, 8.22 \times 10^{-5})$, floored by the artifact's own rounding resolution. Under $\pm 5 \times 10^{-7}$ scan perturbation (worst-case rounding) the sign rule and the scan verdict agree on all 48 cells; at exactly one bisection step ($\pm 6 \times 10^{-7}$) a single boundary cell flips per side, and no strict cell is ever absorbed. The universal $c \leq \min \alpha$ rule is thus the carrier-independent safe budget, and any budget inside the derived margin above α is free extra stopping mass, not a risk. *One geometry* (`discrete_geometry_r506_result.json`). All three non-monotone phenomena of this section are one object, the discrete stopping-state set: $\tau^*(\text{cap})$ is *not* a monotone function (323 of 792 adjacent-cap pairs on the coarse 12-cap grid violate monotonicity; the r501 working note misprinted the pair count), because both scalarizations it reads jump with the certificate-ordered prefix. Yet its full-radius crossing set is always a *prefix*: on the fine grid, every cap at or below c^* certifies the whole simplex and no cap above c^* does (0 holes and 0 overhangs across all 72 cells). The reconciliation is structural, not curve-fitting: shrinking the cap excludes a certificate-ascending prefix, so both scalars $\max_K g_S$ and base_S are monotone in the cap (0 and 0 violations respectively); once $\tau^*{=}1$ both lie at or below α and must stay there, while inside the sub-unit region the two scalarizations disagree and the staircase interior is free to wobble. The deadline's flip sequence is the same object read along the time axis $(0.014, 0.357, 0.214, 0.5, 0.243, 0.5$ as κ shrinks on the probed RLVE atom), and the closed-form edge is the same object read exactly: the hypergeometric flip masses are the quantum, the prefix structure gives the edge c^*, and the slack sign at one step below the closed edge decides its tightness. Reading any of the three as a smooth dial invites exactly the three failures documented above; reading it as a discrete set gives the safe budget, the derived edge, and the two operating ends (cheapest marginal repair at $+0.09\%$–$+1.29\%$ votes; universal repair at $+1.05\%$–$+16.9\%$) of one rule.

Proposition 2 (Prefix law and two-scalar monotonicity of the full-radius crossing set)**.** *Fix a cell $(\widehat{H}_m, \alpha)$ and let $g_{S(c)}$ be the realized profile of the certificate-ordered state-subset rule at flip budget c, with scalarizations $\overline{g}(c) = \max_K g_{S(c)}(K)$ and $\text{base}_S(c) = \sum_K \widehat{H}_m[K]\, g_{S(c)}(K)$, and write $\tau(c) = \tau^*(\widehat{H}_m, g_{S(c)}; \alpha)$. Then:* (i) *both scalarizations are nondecreasing in c;* (ii) *for every c, $\tau(c) = 1$ if and only if $\text{base}_S(c) \leq \alpha$ and $\overline{g}(c) \leq \alpha$;* (iii) *the full-radius crossing set $\{c : \tau(c) = 1\}$ is downward closed — a prefix — while $\tau(c)$ itself is free to be non-monotone inside the sub-unit region; and the crossing out of $\tau{=}1$ as the budget grows is base_S-driven: below α both conditions of* (ii) *already hold at $c = \alpha$ (by domination $\overline{g}(\alpha) \leq \alpha$) and persist for all $c < \alpha$ by* (i)*, so at $c > \alpha$ only the $\text{base}_S(c) \leq \alpha$ condition can bind.*

Proof. (i) With nondecreasing budget, the greedy keep rule retains a nested family of kept sets: every state kept at c' is also kept at $c > c'$ — the certificate-sorted scan reaches that state with at least as much remaining budget, and its own contribution is unchanged. Hence $S(c') \subseteq S(c)$, and since each kept state's flip contribution is nonnegative, both $\overline{g}$ and base_S are nondecreasing. *(ii)* By Proposition 1 the two-sided greedy over B_R adds mass at the highest-g atom and drains it from the lowest-g one, so for $R \in [0,1]$, $V(R) = \mathrm{base}_S(c) + R\,(\overline{g}(c) - \mathrm{base}_S(c))$ — linear in R with slope $\overline{g}(c) - \mathrm{base}_S(c) \geq 0$, hence $\sup_{R\leq 1} V(R) = V(1) = \overline{g}(c)$, and $\sup_{R\leq 1} V(R) \leq \alpha$ holds exactly under the two stated conditions. *(iii)* By *(ii)* the crossing set is an intersection of two sublevel sets of functions nondecreasing in c *(i)*), and a sublevel set of a nondecreasing function is downward closed; the intersection of two prefixes is a prefix. Inside the sub-unit region neither condition of *(ii)* is binding, so τ may wobble there (and does: 323 of 792 adjacent-cap comparisons on the coarse grid); on the fine grid the crossing set has 0 holes and 0 overhangs in all 72 cells, and on the coarse grid the 323 violations decompose into 253 strictly sub-unit wobbles plus 70 upward jumps *at* the prefix edge, with zero holes (`prefix_prop_r508_result.json`). □

F Cross-carrier evidence matrix and unified mechanism

Table 5 collects every in-distribution TEST readout of the paper in one place: the four model–carrier pairs (dependence/resolvable structure decreasing from left to right), all stoppers, all certified levels, and one descriptive α=0.01 CAL-side row for OMR (where the certified family of Section 5 reports only down to α=0.02). Every flip in this matrix is replay agreement with a corresponding full vote and every saving is replay count reduction; it is not gold correctness or operational cost. Validity was certified on CAL ("cert"); TEST is a single descriptive readout under the Bonferroni radius (0.0286 OMR shard 0; 0.0292 shard 1; 0.0310 OpenR1; 0.0307 RLVE). Artifacts: `fit_cal_test_r469`, `shard1_robust_r471` (R1/R2), `openr1_m2_pilot_r473`, `rlve_n8_r474`.

Table 5: Replay evidence matrix: carrier (dependence level) × baseline × endpoint. "null" = no certifiable replay budget at that level; "—" = not applicable / not reported. FULL rows are the zero-count-reduction reference. Seeds 20260815 throughout.

Carrier	rule	α	cert	replay flip	mean k	count reduction
OMR s0 (N=32)	FIXED-HOEF	.10	0.0970	0.0645	7.0	78.1%
	FIXED-EB	.10	0.0958	0.0794	5.0	84.4%
	BAYES-H	.10	0.0495	0.0370	5.1	84.0%
	FIXED-HOEF	.05	0.0471	0.0156	27.0	15.6%
	FIXED-EB	.05	0.0461	0.0324	17.0	46.9%
	BAYES-H	.05	0.0350	0.0249	6.1	80.9%
	FIXED-HOEF	.02	null	—	—	—
	FIXED-EB	.02	0.0171	0.0083	31.0	3.1%
	BAYES-H	.02	0.0163	0.0091	8.5	73.5%
	BAYES-H	.01	0.0119	—	—	—
	FULL-32	—	—	0	32.0	0
OMR s1 (N=32)	FIXED-HOEF	.05	0.0488	0.0171	25.0	21.9%
	FIXED-EB	.05	0.0486	0.0357	15.0	53.1%
	BAYES-H	.05	0.0340	0.0246	6.2	80.6%
	BAYES-H†	.05	0.0341	0.0247	6.2	80.6%
	BAYES-H	.02	0.0158	0.0088	8.6	73.1%
	BAYES-H†	.02	0.0159	0.0089	8.6	73.2%
OpenR1 (M=2)	FIXED-1	—	0.0707	0.0592	1.0	50.0%
	BAYES-H	.10	0.0707	0.0592	1.0	50.0%
	BAYES-H	.05	0.0383	0.0294	1.4	32.1%
	BAYES-H	.02	0.0047	0	2.0	0
RLVE (N=8)	FIXED-HOEF	.10	0.0942	0.0613	5.0	37.5%
	FIXED-EB	.10	0.0783	0.0613	5.0	37.5%
	BAYES-H	.10	0.0469	0.0314	3.4	57.6%
	FIXED-EB/HOEF	.05–.01	null	—	—	—
	BAYES-H	.05	0.0184	0.0103	3.9	50.9%
	BAYES-H	.02	0.0121	0.0046	4.1	48.2%
	BAYES-H	.01	0.0081	0.0019	4.4	45.4%

†Cross-shard prior transfer (prior fit on shard 0 FIT, TV=0.042 to the shard 1 prior); all numbers move $\leq$0.4 points against within-shard. WINDOW3 (no certificate knob) is omitted from the matrix: it has the largest replay count reduction (86.2%) but is invalid at $\alpha \leq 0.05$ (flip 0.0577; Table 1).

Unified mechanism. One quantity — the prior's g-profile — organizes every panel of the paper. (i) *In-distribution*, replay count reduction tracks the resolvable mixture structure: mass on extreme-K problems stops at $k \approx 5$ (OMR, 81% reduction at α=0.05); with only N=8 draws and 74–87% of prior mass on zero-g atoms, three stoppable budgets remain and the count reduction interpolates to 45–58% (RLVE); at M=2 the single stoppable budget yields 32–50% and the certificate degenerates to never-stop exactly when $\max_K g(K) \leq \alpha$ (OpenR1 at α=0.02: count reduction 0, flip 0 — the certificate never over-stops). (ii) *Under prior shift*, the same profile sets the critical radius: τ^* is increasing in the zero-g atom mass across carriers (Spearman 1.0 on carrier means, 16 cells; het-excess variance is not the driver: RLVE $0.0999 < $ OMR 0.129 yet tolerates more), with the within-OMR negative trend (-0.95/-0.89) a staircase artifact (decomposition audit, `spearman_decomposition_r486_result.json`). (iii) *Under synthetic ordered drift*, adaptivity is itself the robustness channel in the specified harness: BAYES-H re-reads the accumulating evidence each draw, so it is the most robust stopper under the smooth E2/E3 mechanisms (valid to δ=0.20 under E2 at α=0.10 where FIXED-k breaks at 0.15), and a margin γ on the certificate buys flip control at a quantified replay-count cost (γ=0.03 covers E2 $\delta \leq 0.20$: flip 0.0491, count reduction 71.8% vs 79.7% unmargined; γ=0.025 covers E3 $\delta \leq 0.15$: flip 0.0495, count reduction 77.2% vs 81.0%; `margin_repair_r469_result.json`). (iv) *The failure boundary is sharp on every axis*: premise-breaking E1 breaks all certificates together by δ=0.10–0.15; on shard 1 at α=0.01 the observed cross-shard mismatch 0.042 exceeds $\tau^*(0.01)$=0.024 and TV alone does not certify the transfer; the coarsest fit m=125 sits outside its ball (TV 0.175) and genuinely breaks at $\alpha \leq 0.02$ (Appendix E(e)). Where the paper claims a win, the margin is large and certified; where it does not, the limiting quantity is named and measured.

G Reproducibility

All numbers trace to JSON artifacts listed in REPRO_README.md; carrier SHA-256 pinned; deterministic seeds; CPU-only; total compute under 25 minutes.